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Efficient construction of effective Hamiltonians with a hybrid machine learning method

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ABSTRACT

The effective Hamiltonian method is a powerful tool for simulating large-scale systems across a wide range of temperatures. However, previous methods for constructing effective Hamiltonian models suffer from key limitations: some require manually predefined interaction terms with limited flexibility in capturing complex systems, while others lack efficiency in selecting optimal interactions. In this work, we introduce the Lasso–GA Hybrid Method (LGHM), a novel approach that combines Lasso regression and genetic algorithms to rapidly construct effective Hamiltonian models. This method is broadly applicable to both magnetic systems (e.g., spin Hamiltonians) and atomic displacement models. To verify the reliability and usefulness of LGHM, we take monolayer CrI_3 and Fe_3GaTe_2 as examples. In both cases, LGHM not only successfully identifies key interaction terms with high fitting accuracy but also reproduces experimental magnetic ground states and Curie temperatures with further Monte Carlo simulations. Notably, our analysis of monolayer Fe_3GaTe_2 reveals that single-ion anisotropy and Heisenberg interaction lead to an out-of-plane ferromagnetic ground state, while fourth-order interactions contribute significantly to the high Curie temperature. Our method is general, so it can be applied to construct other effective Hamiltonian models.

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INTRODUCTION

First-principles calculations have emerged as a crucial tool for studying the structural and electronic properties of solid-state materials.[1,2] Despite its utility, traditional Density Functional Theory (DFT) methods face significant challenges due to their high computational cost, making it difficult to explore finite-temperature properties or handle large-scale systems. Moreover, conventional DFT calculations typically yield ground state properties and final results without directly elucidating the underlying microscopic mechanisms. To overcome these limitations, the effective Hamiltonian method, which is both computationally efficient and capable of delivering deeper physical insights, has gained widespread adoption for tackling thermodynamic and dynamic problems. The effective Hamiltonian method has been applied to investigate various physical systems, including ferroelectric systems and spin systems.[3–5] For instance, effective Hamiltonians of spin models provide insights not only into single-ion anisotropy (SIA), exchange coupling, and Dzyaloshinskii–Moriya interaction (DMI)[6] but also into all possible interaction forms permitted by the system's symmetry.

Our work focuses on deriving the effective Hamiltonian from the DFT calculation, which primarily involves constructing a Hamiltonian model and fitting it to the DFT outcomes.[3,4,7] This process determines the coefficients of various interaction terms, enabling further investigations and analyses. To demonstrate this workflow, let us take the effective Hamiltonian model for magnetic systems as an example. When the interaction terms are truncated to two-body and second-order interactions, the spin component of the Hamiltonian describing the magnetic system can be expressed as $H_{spin} = \sum_{i<j} J_{ij}\vec{S}_i \cdot \vec{S}_j + \sum_{i<j} \vec{D}_{ij} \cdot (\vec{S}_i \times \vec{S}_j) + \sum_i A_i S_{iz}^2$, where the symmetric Heisenberg exchange interactions, antisymmetric Dzyaloshinskii–Moriya interactions, and single-ion anisotropy are considered. Several methods have been developed to estimate the spin interaction parameters,[8–11] for example, the four-state method, where four specific states are calculated using DFT

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to obtain the interaction parameters. This method is among the simplest and most efficient, and it has been extensively utilized in various magnetic systems. However, high-order interactions, such as the biquadratic term,[12] are sometimes important for the physical properties. For instance, the non-Heisenberg behavior in $TbMnO_3$ was successfully explained by the high-order term. When considering the high-order terms, the number of possible interaction terms often becomes large, and least-squares fitting is typically employed to estimate the parameters.[7,13,14] These methods usually require preselecting which interaction terms to be included in the model before performing linear regression. In contrast, machine learning-based methods[7] can automatically identify and retain only the significant interaction terms, streamlining the process and improving the efficiency. This method is particularly advantageous for systems with many possible interactions but few dominant contributions, significantly improving efficiency over conventional methods. However, these advantages diminish in complex systems, such as ferroelectrics and multiferroics, where both the number of possible terms and significant interactions is large, leading to prohibitive computational costs.[15–17]

In this work, we propose a novel fitting approach that combines Lasso regression with genetic algorithms, enabling the rapid construction of effective Hamiltonian models. In the following, we refer to this scheme as the Lasso–GA Hybrid Method (LGHM). It is worth mentioning that this method can be used not only to construct effective Hamiltonian models for magnetic systems but also to construct Hamiltonian models that include atomic displacements. This is possible because we can express the energy as a linear combination of terms that include atomic displacements and strain components.[16,17] We benchmarked our method on a specific dataset and made applications on a model containing both magnetism and atomic displacements (monolayer CrI_3) and a magnetic system model (monolayer Fe_3GaTe_2), demonstrating its effectiveness. In the monolayer Fe_3GaTe_2, we further conducted an in-depth study of its higher-order interaction terms and their impact on the magnetic ground state and phase transition temperature, identifying the factors contributing to its high Curie temperature.

METHOD

Effective Hamiltonians are typically expressed as linear combinations of specific operators or interaction terms. The general form of effective Hamiltonians can be written as

$$H_{\text{eff}} = \sum_{j=1}^{p_{\max}} C_j h_j, \tag{1}$$

where h_j are specific operators or interaction terms, C_j are the corresponding coefficients, and $p_{\max}$ is the number of all possible terms. The specific operators can be obtained by Property Analysis and Simulation Package for materials (PASP).[18] Determining the coefficients C_j appears to be a straightforward linear regression problem, which involves minimizing the sum of squares as shown in the following equation, where y_i is the exact value and $\widehat{y_i}$ is the value predicted by the linear model,

$$\sigma^2 = \frac{\sum_{i=1}^{n} (y_i - \widehat{y_i})^2}{n}. \tag{2}$$

However, there are two potential challenges when performing direct linear fitting. The first challenge is the limited number of data points. Linear regression requires the number of data points to exceed the number of parameters. Since the data points are obtained through DFT calculations, a large number of interaction terms would necessitate more DFT computations, resulting in higher computational costs. The second challenge is that even with sufficient data, too many interaction terms may lead to overfitting, reducing the model's predictive accuracy. To address the above challenges, we need a model that is accurate and incorporates as few interaction terms as possible.

Our method combines Lasso regression[19,20] and genetic algorithms.[21] By defining the optimization problem and the genetic representation properly, the genetic algorithm can help identify the relevant interaction terms.

Inspired by MLMCH,[7] we adopt the so-called L_0 regularization method. Similar to L_1 and L_2 regularization techniques commonly used in machine learning, we reformulate the linear model fitting task, which is minimizing the sum of squares as shown before, into the minimization of Eq. (3), where λ is a hyperparameter and p is the number of selected terms. λ^p is a penalty term. Minimizing s allows us to balance both the accuracy of the model and the reduction in the number of model parameters,

$$s = \sigma^2 \times \lambda^p. \tag{3}$$

In the genetic algorithm, we can represent the inclusion or exclusion of a specific interaction term using 1 and 0, respectively. The set of all selected terms in the model can thus be encoded as a binary gene consisting of 0s and 1s. The optimization problem is formulated as minimizing the s mentioned above. However, in cases where the number of potential interaction terms reaches thousands, directly applying the genetic algorithm would be highly time-consuming. Therefore, our approach first employs Lasso regression, which can reduce small coefficients to zero,[20] to pre-select relevant interaction terms before using the genetic algorithm.

The complete workflow of our LGHM is shown in Fig. 1. First, we perform Lasso regression as a preliminary fitting and initial selection of interaction terms. Then, based on the selected terms, we proceed with further filtering using a genetic algorithm. In particular, we initialize a set of genes, each representing whether a particular term is retained. We then perform linear regression on these genes and calculate their loss values. If the best gene satisfies certain conditions, such as no update in the best loss over 40 consecutive rounds, the loop exits; otherwise, the next evolutionary round begins. We select a subset of genes with relatively small loss values from all the results. These genes then undergo crossover and mutation to form new genes, and the process loops back to the linear regression and loss calculation step.

BENCHMARK

We benchmark LGHM on a specific dataset designed to simulate the selection and fitting of relevant interaction terms from a pool of 6000 possible interactions. In this setup, 1000 interaction coefficients are randomly assigned non-zero values, with their magnitudes generated within a specified range. This range decreases progressively, simulating a realistic situation where nearest-neighbor,

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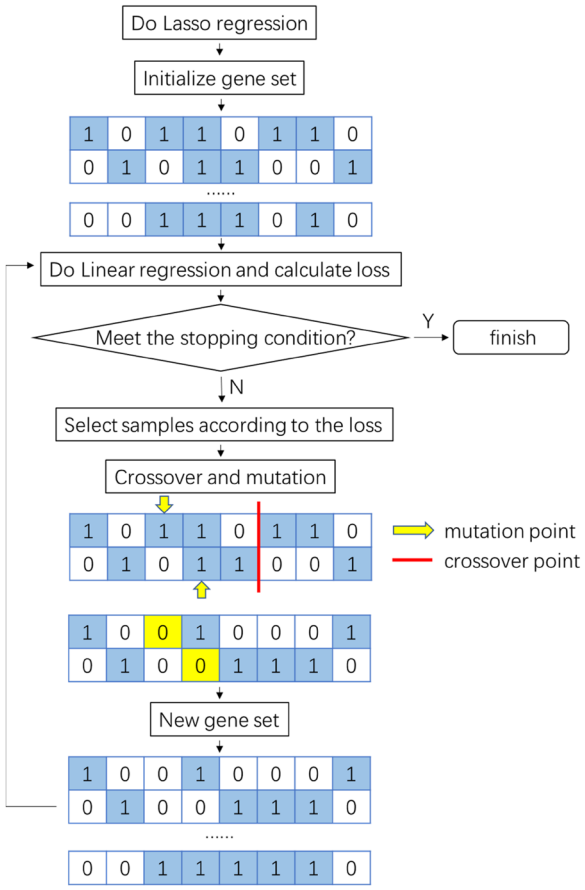


FIG. 1. Flow chart of LGHM. The yellow arrows point out the mutation point, and the red line means the crossover point.

low-order interactions are larger, while higher-order, long-range interactions are smaller. The invariants are similarly generated at random. Subsequently, matrix multiplication is performed on the selected interactions, followed by the addition of Gaussian noise to simulate experimental uncertainty. This specific dataset contains a total of 5000 data points, with 4000 used for model fitting and the remaining 1000 reserved as the test set. LGHM takes 37 604 s on two GPUs and 286 910 s on a CPU to get the fitting results of ten λ, while the previous method[7] fails to give the results even after running for two weeks.

The fitting results with $\lambda = 1.003$ are presented below. Our method produces a model with 668 interaction terms, achieving a mean absolute error (MAE) of 0.002 12 and an R^2 value of 0.9999. The parity plot of the model on the test dataset is shown in Fig. 2. Among the coefficients we manually set, 601 terms have absolute values greater than 0.001, all of which are successfully identified by our method. In the supplementary material, we provide the complete list of interaction terms with absolute values greater than 1.

This specific dataset demonstrates the capability of LGHM to efficiently identify significant interaction terms even when the total number of possible terms is extremely large. In addition, it shows that the method remains effective even when the number of data points is smaller than the number of possible interaction terms.

APPLICATION ON MONOLAYER CrI_3

Significant progress has recently been made in 2D magnets, making it possible to further push spintronic devices to the atomic level.[22,23] Among these, monolayer CrI_3 was reported as a 2D material with an out-of-plane ferromagnetic ground state.[24,25] This material exhibits a significant magnetic anisotropy, with a Curie temperature of 45 K.[26] In order to get more insights into the microscopic magnetic properties of CrI_3, we develop its first-principles-based effective Hamiltonians. Here, we adopt the symmetry-adapted cluster expansion method, as implemented in PASP.[18] Such a method roots in cluster expansion that goes over all combinations of spin components and atomic displacements, enabling the generation of all possible forms of interactions, i.e., the invariants. The initial model contains enough invariants, including interactions up to the four-body and fourth-order (the truncation distances of the second-, third-, and fourth-order terms are set to 12, 8, and 5 bohr). After symmetry analyses, the number of possible interaction terms (p_{max}) is 25 448. To fit and refine the coefficients of these invariants, a dataset of 5017 data points is generated using DFT calculations (the calculation details can be found in the supplementary material), where 4000 data points are used for fitting and 1017 data points are used for testing. In this dataset, each data point represents a structure consisting of 72 atoms, where both displacement degrees of freedom and magnetic state degrees of freedom are allowed.

Our method completes the fitting process of ten λ in 79 983 s on two GPUs and 327 725 s on a CPU. The result corresponding to the lowest mean squared error ($\lambda = 1.0009$) is selected. This yields a model with 737 interaction terms, achieving a mean absolute error (MAE) of 0.002 35 eV/u.c. and an R^2 value of 0.9997. The parity plot of the model on the test dataset is shown in Fig. 3(b). Using the model obtained from our method, we perform Monte Carlo simulations and conjugate gradient (CG) optimizations to determine the ground state and phase transition temperature of monolayer CrI_3. As shown in Fig. 3(a), monolayer CrI_3 exhibits an out-of-plane ferromagnetic ground state with a phase transition temperature of 25 K, in line with previous studies.[24–26] The details of the dataset and Monte Carlo simulations can be found in the supplementary material.

APPLICATION ON MONOLAYER Fe_3GaTe_2

Fe_3GaTe_2 (Fig. 4) has emerged as a promising candidate material for next-generation spintronic devices due to its robust ferromagnetic properties down to the monolayer limit. Recent studies have highlighted its strong ferromagnetism, with notable findings indicating a Curie temperature ranging from 350 to 380 K.[27–30] Given these properties, our work aims to construct an exact spin Hamiltonian of Fe_3GaTe_2 and explore the microscopic origin of magnetic properties. To achieve this, we employed the PASP software to generate all possible invariants. The truncation distances of the second-, third-, and fourth-order terms are set to 14, 8.8, and 10 bohr, which is up to the 10th, 11th, and 21st nearest pairs, respectively. After symmetry analyses, the number of possible interaction terms (p_{max}) is 259. To fit and refine the coefficients of these

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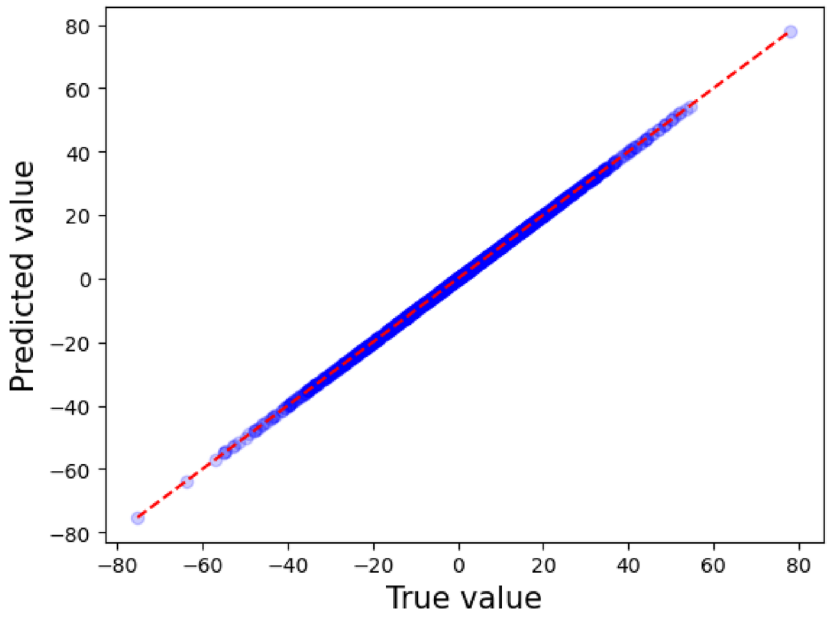


FIG. 2. A plot of predicted values vs true values. Different blue dots indicate different samples in the test set.

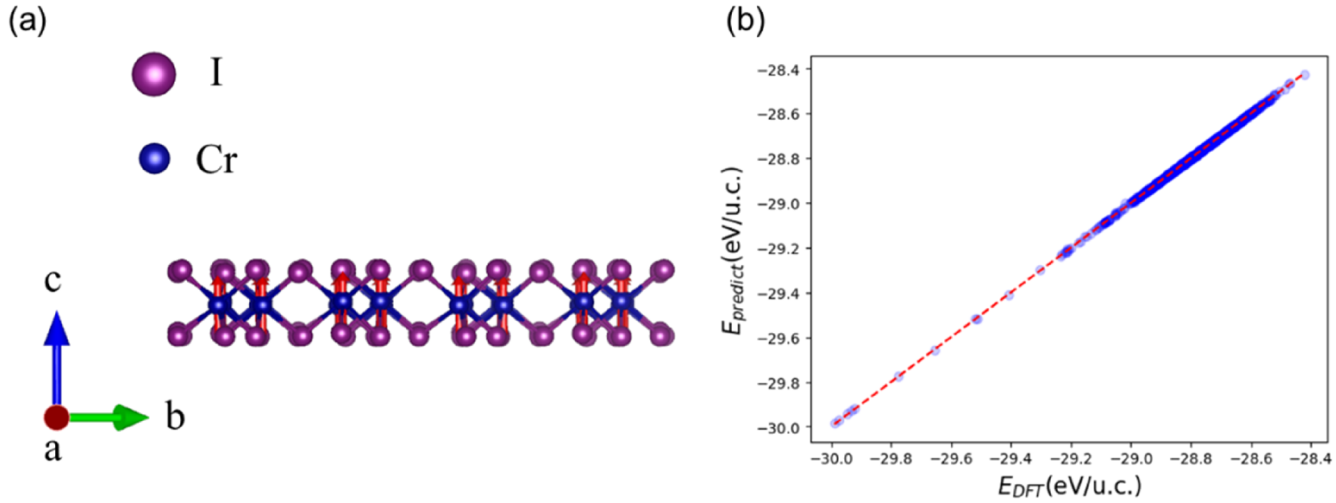


FIG. 3. (a) Ground state of monolayer CrI_3. (b) A plot of predicted energies (per unit cell) vs computed values (using *ab initio* calculations).

invariants, a dataset of 1344 data points is generated using DFT calculations, where 1049 data points are used for fitting and 295 data points are used for testing.

It takes 774 s for our method on a CPU to get the fitting results of ten λ. The result of the lowest mean squared error ($\lambda = 1.008$) is selected. Our method produces a model with 47 interaction terms, achieving a mean absolute error (MAE) of 0.000 99 eV/u.c. and an R^2 value of 0.9998. Among the 47 interaction terms, the 14 most significant ones are listed in Table I. We perform a MC simulation and CG optimization on our model to investigate the ground state and the phase transition on Fe_3GaTe_2. The simulation reveals

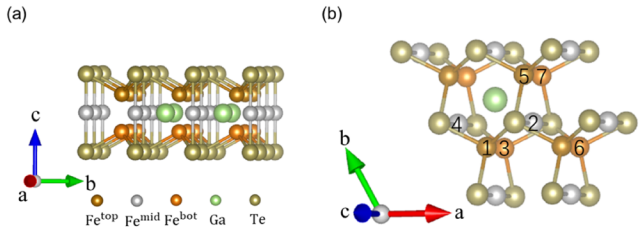


FIG. 4. (a) Structure of monolayer Fe_3GaTe_2. (b) The numbering of the Fe atoms corresponds to the subscripts of the magnetic moments in Table I.

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TABLE I. Selected magnetic parameters of monolayer Fe_3GaTe_2. The superscripts "*t*," "*b*," and "*m*" represent the Fe atoms located in the top, bottom, and middle layer, respectively, which is illustrated in Fig. 4(a). *A* stands for single-ion anisotropy, *J* stands for exchange couplings, *D* stands for DMI, *B* stands for biquadratic interaction, and *K* stands for fourth-order coefficients. The subscripts "1" and "2" refer to the atoms indicated in Fig. 4(b).

Index	Symbol	Value	Invariant	Index	Symbol	Value	Invariant
1	A^t_{zz}	−2.0	S_z^2	8	B^{tb}	−7.4	$\left(\vec{S_1}\cdot\vec{S_3}\right)^2$
2	A^m_{zz}	1.4	S_z^2	9	K_1	−5.7	$(\vec{S_3}\cdot\vec{S_1})\cdot(\vec{S_2}\cdot\vec{S_3})$
3	J^{tb1}	6.4	$\vec{S_1}\cdot\vec{S_3}$	10	K_2	−5.7	$(\vec{S_3}\cdot\vec{S_1})\cdot(\vec{S_2}\cdot\vec{S_1})$
4	J^{tm}	−11.2	$\vec{S_1}\cdot\vec{S_2}$	11	K_3	−5.1	$(\vec{S_1}\cdot\vec{S_2})\cdot(\vec{S_2}\cdot\vec{S_3})$
5	J^{tt}	4.1	$\vec{S_1}\cdot\vec{S_5}$	12	K_4	−5.3	$(\vec{S_2}\cdot\vec{S_3})\cdot(\vec{S_2}\cdot\vec{S_6})$
6	D^{tt}	−5.3	$S_{1x}S_{5y}-S_{5x}S_{1y}$	13	K_5	−6.3	$(\vec{S_2}\cdot\vec{S_3})\cdot(\vec{S_2}\cdot\vec{S_5})$
7	J^{tb2}	−6.4	$\vec{S_3}\cdot\vec{S_5}$	14	K_6	−4.6	$(\vec{S_3}\cdot\vec{S_7})\cdot(\vec{S_1}\cdot\vec{S_5})$

that the ground state is out-of-plane ferromagnetic, which agrees well with the experiment,[27] with the Curie temperature being 460 K (full model in Fig. 5). The details of the dataset and Monte Carlo simulations can be found in the supplementary material.

Although the ground state of 2D Fe_3GaTe_2 is out-of-plane ferromagnetic, as indicated by both experimental results and our simulations, its microscopic mechanism remains unclear. According to our calculations, as shown in Table I, the nearest-neighbor Heisenberg interaction exhibits an antiferromagnetic tendency. This is in contrast to previous studies,[28,31] whose results suggest that the nearest-neighbor Heisenberg interaction is ferromagnetic. This discrepancy may arise from differences in the employed models. In our study, numerous high-order interactions are considered, particularly the biquadratic interactions, which favor the ferromagnetic state and have a non-negligible magnitude.

To illustrate the significance of those interaction terms, MC simulations and CG optimizations were performed on two additional models: model A, incorporating the 14 interaction terms listed in Table I, and model B, considering only the single-ion anisotropy and exchange coupling mentioned in Table I. Both simulations reveal an out-of-plane ferromagnetic ground state. Model A indicates a Curie temperature of 350 K, which is close to that of the

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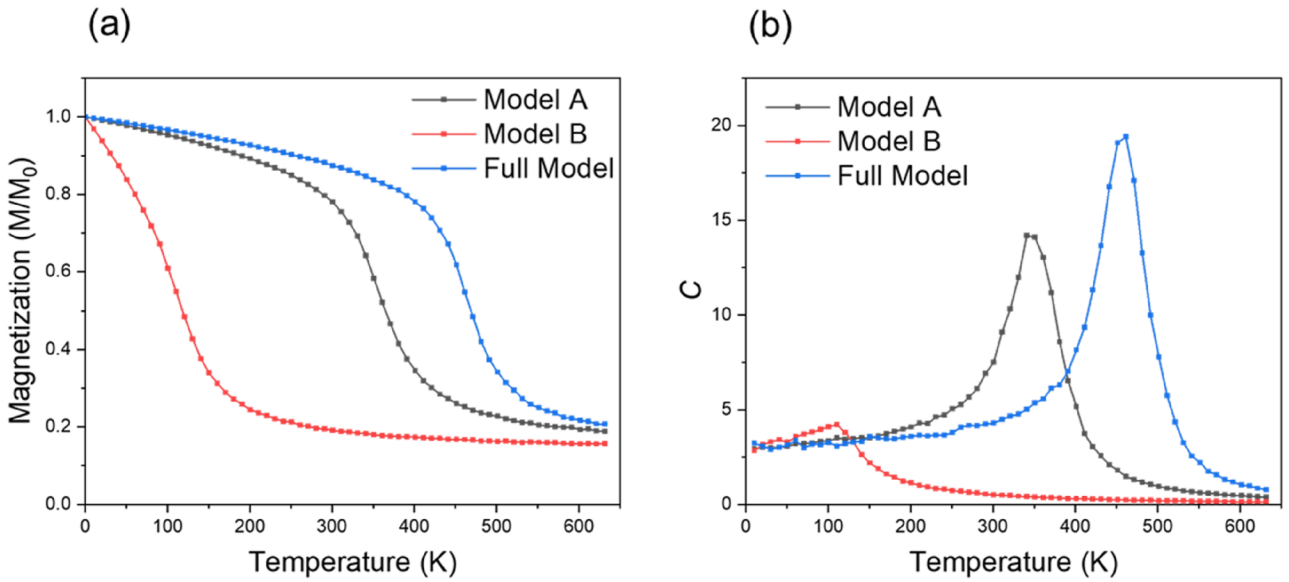


FIG. 5. (a) Average magnetization of Fe atoms at different temperatures of monolayer Fe_3GaTe_2. Lines of different colors represent different models. Model A contains only the A and J interactions listed in Table I, and model B contains all the interactions listed in Table I, while the full model refers to the model with all the 47 interactions found by our method.

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TABLE II. Energy difference per atom between different magnetic states of monolayer FGT in units of meV/atom. OP and IP stand for out of plane and in plane, while FM and FIM stand for ferromagnetic and ferrimagnetic. Special stands for a specific spin state, which is shown in the supplementary material. Full model refers to the model with all the interactions found by the LGHM. Model A refers to the model with only the *A* and *J* interactions listed in Table I. Model B refers to the model with all the interactions listed in Table I. The spin configurations of the magnetic states are shown in the supplementary material.

Method \ Magnetic state	OP-FM	IP-FM	OP-FIM	Special
DFT	0	0.15	66.79	42.54
Full model	0	0.15	66.69	41.44
Model A	0	0.43	21.84	11.25
Model B	0	0.43	54.55	35.19

full model, while model B yields a Curie temperature of 120 K. The significant difference in Curie temperatures between these models implies the crucial role of higher-order interactions. This is understandable since the magnitudes of higher-order interactions are comparable to those of Heisenberg exchange interactions, and their cooperative effect tends to favor the ferromagnetic state. Table II presents the energy disparities between different magnetic states calculated using DFT, the full model, and the model with selected terms, further demonstrating the energy contribution of high-order interactions. In contrast to Fe_3GeTe_2,[3,32] whose Curie temperature primarily originates from strong Heisenberg exchange interactions, Fe_3GaTe_2 exhibits a higher Curie temperature, which can be mainly attributed to its fourth-order interactions. This finding significantly deepens the understanding of high-order interaction-driven high Curie temperatures and highlights the potential of Fe_3GaTe_2 in advanced spintronic applications.

In conclusion, we proposed the LGHM for obtaining effective Hamiltonian models. We benchmarked this method on a specific dataset, demonstrating its effectiveness. Subsequently, we applied our approach to monolayer CrI_3 and Fe_3GaTe_2, where we identified the interaction terms and performed Monte Carlo simulations using these terms. The results yield reasonable magnetic ground states and Curie temperatures, further validating the robustness of our method. Notably, in the case of monolayer Fe_3GaTe_2, we found that the higher-order interactions are crucial for its high Curie temperature.

Compared to the previous MLMCH method, our approach is more efficient in obtaining results when dealing with a large number of possible interaction terms, for the sake of the initial selection performed using L1 regularization before genetic algorithm. This is particularly common when the model considers atomic displacements, making LGHM especially suitable for such complex scenarios.

SUPPLEMENTARY MATERIAL

See the supplementary material for more details of the dataset and Monte Carlo simulations, which includes Refs. 2, 19–21, and 33–38.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yang Cheng and Binhua Zhang contributed equally to this work.

Yang Cheng: Data curation (equal); Investigation (lead); Methodology (lead); Visualization (equal); Writing – original draft (equal). **Binhua Zhang**: Data curation (equal); Investigation (equal); Methodology (equal); Visualization (equal); Writing – original draft (equal). **Xueyang Li**: Data curation (equal); Methodology (equal). **Hongyu Yu**: Data curation (equal); Methodology (equal). **Changsong Xu**: Conceptualization (equal); Funding acquisition (equal); Supervision (equal); Writing – review & editing (equal). **Hongjun Xiang**: Conceptualization (equal); Funding acquisition (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article and its supplementary material.

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